

PHYSICS CLASS 9

Complete Subjective Solutions

FBISE SSC-I | 2nd Annual 2025

Sections B and C - every question and every OR option

Prepared for: Fakhar STEM Sphere

Coverage: Q2(i-xi), both alternatives; Q3-Q6, both alternatives.

Paper marks: Section B: 33; Section C: 20.

These model answers use simple English, essential scientific terms and complete numerical steps. They are designed to address the marks shown on the supplied paper; awarded marks depend on the examiner and the answer written.

Numerical convention: Use $g = 10 \text{ m/s}^2$, as printed on the paper. Write given data, the required quantity, the formula, substitution and the final answer with its unit.

How to revise: Learn the bold definitions and equations, practise the numerical steps, and redraw the labelled diagrams. Both alternatives are solved here for preparation. In the actual paper, follow the OR instructions and attempt the required alternative.

Source note: Solutions follow the questions in the attached subjective-paper scan. The year in this document follows the uploaded filename.

SECTION B | Q2(i) and Q2(ii)

Q2(i) - What is force? State its SI unit. [2 + 1]

Force is a push or pull that can change, or tend to change, the state of rest or motion of a body, its direction of motion, or its shape.

Its SI unit is the **newton (N)**. One newton produces an acceleration of 1 m/s^2 in a mass of 1 kg :

$$1 \text{ N} = 1 \text{ kg m/s}^2.$$

OR - Justify that displacement is a vector quantity. Draw a diagram. [2 + 1]

Displacement is the directed change in position from the initial point to the final point. Its magnitude is the shortest straight-line distance between these points, and its direction is from the initial point towards the final point. Since it has **both magnitude and direction** and follows vector addition, it is a vector quantity. For example, 5 m east specifies displacement; 5 m alone does not specify it completely.



Q2(ii) - State and briefly explain Newton's first law. [1 + 2]

Law: A body remains at rest or continues to move with uniform velocity in a straight line unless acted upon by a net external force.

Explanation: When the resultant force is zero, a body does not change its velocity. The tendency to resist a change in the state of motion is called **inertia**. For example, a book remains at rest on a table until an unbalanced force moves it. A moving object slows because friction or another external force acts on it.

OR - Compare scalar and vector quantities: any three points. [3]

Scalar quantities	Vector quantities
Have magnitude only.	Have magnitude and direction.
Are added by ordinary arithmetic.	Are added using vector addition, accounting for direction.
Examples: mass, time, distance and speed.	Examples: force, displacement, velocity and acceleration.

SECTION B | Q2(iii) and Q2(iv)

Q2(iii) - Write any three methods to reduce friction. [1 × 3]

- 1. Lubrication:** Apply oil or grease between moving surfaces. The lubricant separates them and reduces direct contact.
- 2. Polishing:** Make surfaces smoother so that their irregularities interlock less.
- 3. Ball bearings or rollers:** Replace sliding friction with rolling friction, which is generally much smaller.

OR - How many kilometres are there in 25 micrometres? [3]

Given: Length = 25 μm .

1 μm = 10^{-6} m and 1 m = 10^{-3} km.

Therefore, 1 μm = 10^{-9} km.

25 μm = 25×10^{-9} km = **2.5×10^{-8} km.**

Q2(iv) - Differentiate between manometer and barometer. [3]

Manometer	Barometer
Measures the pressure of a gas in a vessel, often relative to atmospheric pressure.	Measures atmospheric pressure.
A common form is a U-shaped tube partly filled with liquid, with one end connected to the gas.	A mercury barometer has a long tube closed at the top and inverted into a mercury reservoir.
The difference in liquid levels gives the pressure difference: $\Delta P = \rho g \Delta h$.	The height of the mercury column gives atmospheric pressure: $P = \rho gh$.

OR - Define density. Write its formula and unit. [1 × 3]

Density is the mass per unit volume of a substance.

Formula: $\rho = m/V$, where m is mass and V is volume.

SI unit: kilogram per cubic metre, **kg/m^3** . Another commonly used unit is g/cm^3 .

SECTION B | Q2(v) and Q2(vi)

Q2(v) - Compare permanent magnets and electromagnets. [3]

Permanent magnet	Electromagnet
Retains its magnetism without an electric current.	Produces a magnetic field when current flows through its coil; a soft-iron core greatly strengthens it.
Cannot normally be switched on and off.	Can be switched on and off by controlling the current.
Its strength and polarity are not readily changed during normal use.	Its strength can be varied by changing current; reversing current reverses its poles.

OR - Describe the process of magnetic recording. [3]

1. Sound or other information is converted into a changing electrical signal.
2. This signal passes through the coil of a recording head, producing a changing magnetic field.
3. As a magnetic tape or disk moves past the head, its small magnetic regions become magnetised according to the signal. The pattern remains stored and can later be read by a playback head to reproduce the information.

Q2(vi) - Describe the effect of temperature on particle motion. [3]

Temperature is related to the **average kinetic energy** of particles. When temperature increases, particles gain average kinetic energy. Particles in liquids and gases move faster, while particles in a solid vibrate more vigorously about their fixed positions. On cooling, their average kinetic energy decreases, so their movement or vibration becomes less vigorous.

OR - Describe the working of a hydraulic brake system. [3]

A hydraulic brake works on **Pascal's principle**: pressure applied to an enclosed liquid is transmitted equally in all directions.

1. Pressing the brake pedal pushes the piston in the master cylinder and increases pressure in the brake fluid.
2. The fluid transmits this pressure through pipes to the wheel cylinders or brake calipers. Since $F = PA$, the pressure produces a force on their pistons.
3. These pistons press brake shoes against drums or pads against discs. Friction opposes wheel rotation and slows or stops the vehicle.

SECTION B | Q2(vii) and Q2(viii)

Q2(vii) - Show that micro is a thousand times smaller than milli. [3]

Milli (m) = 10^{-3} of a unit; **micro (μ)** = 10^{-6} of a unit.

micro/milli = $10^{-6}/10^{-3} = 10^{-3} = 1/1000$.

Thus, the micro prefix represents **one-thousandth** of the milli prefix.

Example: **1 mm = 1000 μ m**, or $1 \mu\text{m} = 0.001 \text{ mm}$.

OR - Differentiate between classical mechanics and quantum mechanics. [3]

Classical mechanics	Quantum mechanics
Describes ordinary macroscopic bodies, such as balls, vehicles and planets.	Describes matter at atomic and subatomic scales, such as electrons and atoms.
Uses laws such as Newton's laws to describe motion.	Uses quantum laws; quantities such as the energy of bound electrons have discrete allowed values.
An ideal particle has a definite trajectory determined from initial conditions and forces.	Predicts probabilities for measurement outcomes; position and momentum cannot both be specified with unlimited precision.

Q2(viii) - Briefly describe any three sub-fields of physics. [3]

1. Mechanics: The study of motion, forces and their effects on bodies.

2. Thermodynamics: The study of heat, temperature and energy transfer or conversion.

3. Optics: The study of light and phenomena such as reflection and refraction.

OR - Derive the formula for liquid pressure. [3]

Consider a stationary liquid column of height h , cross-sectional area A and density ρ .

Volume: $V = Ah$.

Mass: $m = \rho V = \rho Ah$.

Weight: $F = mg = \rho Ahg$.

Pressure due to the liquid column: $P = F/A = \rho Ahg/A$.

Therefore, $P = \rho gh$.

Here h is the vertical depth below the liquid surface. If the surface is exposed to air, total pressure is $P_{\text{total}} = P_{\text{atm}} + \rho gh$.

SECTION B | Q2(ix) and Q2(x)

Q2(ix) - Differentiate between magnetic and nonmagnetic materials: three points. [3]

Magnetic materials	Nonmagnetic materials
Are strongly attracted by an ordinary magnet.	Are not appreciably attracted by an ordinary magnet.
Can be strongly magnetised, for example by a magnetic field.	Cannot be made into ordinary strong magnets.
Examples: iron, nickel and cobalt.	Examples: wood, plastic and glass.

Concept note: “Nonmagnetic” here is the school-level term for materials without a readily noticeable magnetic response.

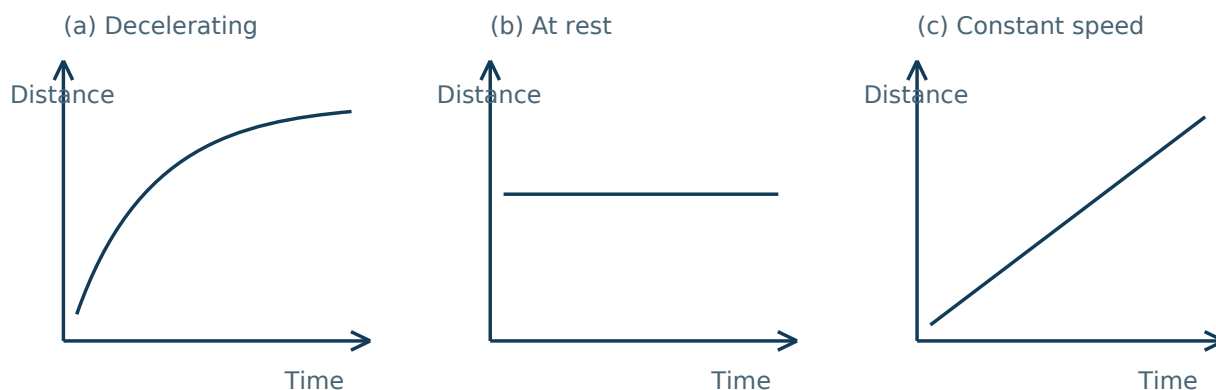
OR - Can we drink water with a straw on the Moon? Justify. [1 + 2]

No, not in the Moon’s open environment. When we suck through a straw on Earth, pressure inside the straw decreases. Atmospheric pressure acting on the water’s surface then pushes the water up the straw. The Moon has virtually no atmosphere, so there is no appreciable atmospheric pressure to push water up in this way.

Clarification: A straw can work inside a pressurised lunar habitat, where air pressure is present.

SECTION B | Q2(x) - Graphs and pressure

Q2(x) - Draw distance-time graphs for the following motions. [3]



(a) Decelerating: Distance increases, but the slope becomes smaller with time because speed decreases.

(b) At rest: A horizontal line shows that distance remains constant.

(c) Constant speed: A straight rising line has constant slope.

Remember: Speed is the gradient of a distance-time graph. A decelerating body's distance does not decrease; the rate of increase becomes smaller.

OR - Find the pressure on a submarine at a depth of 6 km. [3]

Given: $h = 6 \text{ km} = 6000 \text{ m}$; $\rho = 1000 \text{ kg/m}^3$; $g = 10 \text{ m/s}^2$.

Required: Pressure due to the water, P .

Formula: $P = \rho gh$.

Substitution: $P = 1000 \times 10 \times 6000$.

Answer: $P = 6.0 \times 10^7 \text{ Pa} = 60 \text{ MPa}$.

This is the water pressure (gauge pressure). Absolute pressure would also include the pressure at the ocean surface; the school numerical uses $P = \rho gh$.

SECTION B | Q2(xi)

Q2(xi) - What is a theory? How is it developed? [1 + 2]

Definition: A scientific theory is a well-supported explanation of natural phenomena based on evidence.

Development: Scientists make observations, identify a problem and propose a hypothesis. They use the hypothesis to make predictions and test them through experiments or further observations. If repeated independent tests support the explanation over a wide range of observations, it can become part of an accepted theory. A theory may be refined or revised when new evidence is obtained.

OR - Express the following using prefixes. [3]

(a) $29.3 \times 10^{-10} \text{ m}$

$$29.3 \times 10^{-10} \text{ m} = 2.93 \times 10^{-9} \text{ m.}$$

Since $10^{-9} \text{ m} = 1 \text{ nm}$, the answer is **2.93 nm** (nanometres).

(b) $0.00034 \times 10^{-2} \text{ s}$

$$0.00034 = 3.4 \times 10^{-4}.$$

$$0.00034 \times 10^{-2} \text{ s} = 3.4 \times 10^{-6} \text{ s.}$$

Since $10^{-6} \text{ s} = 1 \mu\text{s}$, the answer is **3.4 μs** (microseconds).

Section B - quick numerical check

Question	Final answer
Q2(iii), OR	$25 \mu\text{m} = 2.5 \times 10^{-8} \text{ km}$
Q2(vii), main	micro/milli = 1/1000
Q2(x), OR	$P = 6.0 \times 10^7 \text{ Pa}$
Q2(xi), OR (a)	2.93 nm
Q2(xi), OR (b)	3.4 μs

SECTION C | Q3 - Orbital motion [1 + 4]

Define orbital motion and derive the average orbital speed of a satellite.

Orbital motion: The motion of a body around another body along an orbit is called orbital motion. For example, an artificial satellite revolves around Earth under the action of gravity.

Derivation:

Consider a satellite travelling in a circular orbit of radius r . Let T be its orbital period, the time taken to complete one revolution.

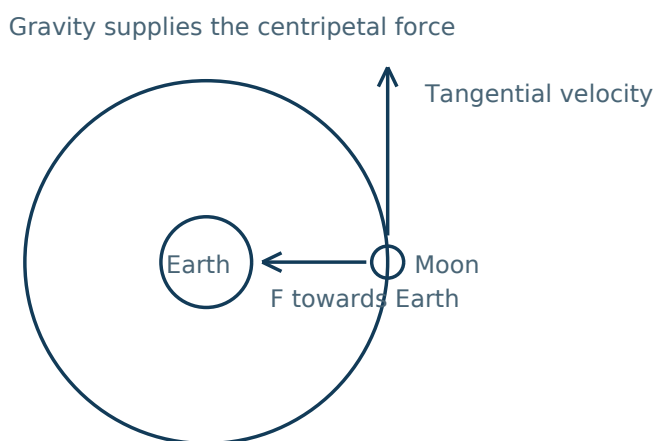
1. Distance travelled in one revolution = circumference of the orbit = $2\pi r$.

2. Average speed = total distance / total time.

3. Therefore, $v = 2\pi r/T$.

4. If the satellite is at height h above Earth's surface, $r = R + h$, where R is Earth's radius. Hence, $v = 2\pi(R + h)/T$.

SI units: r in metres, T in seconds and v in m/s.



For a satellite: Replace “Moon” in the diagram with “satellite.” The orbital radius is measured from Earth's centre, not its surface. Gravity acts towards Earth's centre, while instantaneous velocity is tangential to the orbit.

Additional relation for a circular gravitational orbit: If the mass M of Earth is given, gravity supplies centripetal force:

$$GMm/r^2 = mv^2/r \Rightarrow v^2 = GM/r \Rightarrow v = \sqrt{GM/r}.$$

For the requested average orbital speed, the main derivation is $v = 2\pi r/T$. The circular-orbit assumption is essential; a noncircular orbit has a different path length.

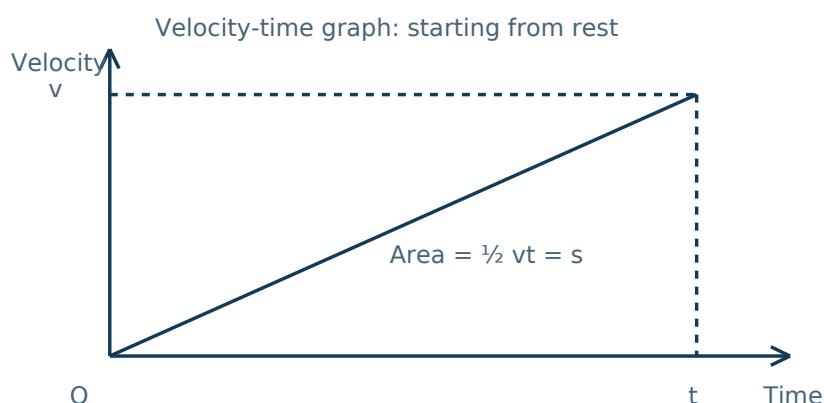
SECTION C | Q3 - OR: Kinetic energy [1 + 4]

Define kinetic energy and derive its expression using a graphical method.

Kinetic energy is the energy possessed by a body due to its motion.

Derivation using a velocity-time graph:

Consider a body of mass m initially at rest. A constant net force F produces uniform acceleration a , and its speed becomes v after time t . Ignore dissipative losses.



1. Find displacement from the graph.

The area under the velocity-time graph equals displacement. The area is a triangle:

$$s = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2}tv.$$

2. Apply Newton's second law.

Since initial speed is zero, $a = (v - 0)/t = v/t$.

Therefore, $F = ma = mv/t$.

3. Calculate the work done.

$$W = Fs = (mv/t)(\frac{1}{2}tv) = \frac{1}{2}mv^2.$$

4. Relate work to kinetic energy.

The work done becomes the body's kinetic energy. Therefore,

$$\text{K.E.} = \frac{1}{2}mv^2.$$

Here m is in kg and v is in m/s; kinetic energy is in joules (J).

Exam tip: Draw and label both axes, show the triangular area, and write the steps connecting s , F , W and K.E. If initial speed is u rather than zero, the same method gives $W = \frac{1}{2}m(v^2 - u^2)$.

SECTION C | Q4 - Both numerical alternatives [5 each]

Q4 - A 3 kg block is dropped and reaches the ground at 80 m/s. Find its time of fall and the height.

Given: $m = 3 \text{ kg}$; initial velocity $u = 0$ (dropped from rest); final velocity $v = 80 \text{ m/s}$; acceleration $g = 10 \text{ m/s}^2$ downward.

Required: Time t and height h . Take downward as positive; ignore air resistance.

Step 1: Find the time.

$$v = u + gt.$$

$$t = (v - u)/g = (80 - 0)/10.$$

$$\mathbf{t = 8 \text{ s.}}$$

Step 2: Find the height.

$$v^2 = u^2 + 2gh.$$

$$h = (v^2 - u^2)/(2g) = (80^2 - 0)/(2 \times 10).$$

$$h = 6400/20.$$

$$\mathbf{h = 320 \text{ m.}}$$

Check: $h = ut + \frac{1}{2}gt^2 = 0 + \frac{1}{2} \times 10 \times 8^2 = 320 \text{ m.}$

Final answers: Time = **8 s**; height = **320 m**.

The mass does not enter these free-fall calculations because all bodies have the same gravitational acceleration when air resistance is neglected.

OR - Find the effort to lift a 2500 kg car using a hydraulic jack. Small piston: 6 cm²; large piston: 90 cm².

Given: Car mass $m = 2500 \text{ kg}$; $A_1 = 6 \text{ cm}^2 = 6 \times 10^{-4} \text{ m}^2$; $A_2 = 90 \text{ cm}^2 = 9 \times 10^{-3} \text{ m}^2$; $g = 10 \text{ m/s}^2$.

Required: Force F_1 applied to the small piston.

Step 1: Find the load force.

$$F_2 = mg = 2500 \times 10 = \mathbf{25,000 \text{ N.}}$$

Step 2: Apply Pascal's principle.

$$F_1/A_1 = F_2/A_2.$$

$$F_1 = F_2(A_1/A_2) = 25,000 \times (6/90).$$

$$\mathbf{F_1 = 1666.7 \text{ N} \approx 1.67 \times 10^3 \text{ N.}}$$

Final answer: The ideal force needed at the small piston is approximately **1667 N**.

Exam tip: The question gives 2500 kg, which is mass. Convert it to weight in newtons before applying the pressure formula. Since both areas are in cm², their ratio can be used directly.

SECTION C | Q5 - Stopping a car and a van [2 + 2 + 1]

Which is more difficult to stop in 5 seconds: a 2500 kg car at 10 m/s, or a 5000 kg van at 4 m/s?

Principle: The average stopping force is the rate of change of momentum: $F = \Delta p / \Delta t$. Both vehicles must reach $v = 0$ in $t = 5$ s.

(a) Car:

$m = 2500$ kg; $u = 10$ m/s; $v = 0$; $t = 5$ s.

Initial momentum: $p_i = mu = 2500 \times 10 = 25,000$ kg m/s.

Final momentum: $p_f = 0$.

$F = m(v - u)/t = 2500(0 - 10)/5 = -5000$ N.

Magnitude of stopping force = 5000 N.

(b) Van:

$m = 5000$ kg; $u = 4$ m/s; $v = 0$; $t = 5$ s.

Initial momentum: $p_i = mu = 5000 \times 4 = 20,000$ kg m/s.

$F = m(v - u)/t = 5000(0 - 4)/5 = -4000$ N.

Magnitude of stopping force = 4000 N.

Conclusion: The **car is more difficult to stop in 5 seconds** because its initial momentum is greater and it requires a larger average stopping force: 5000 N compared with 4000 N for the van. The negative signs show that stopping forces act opposite to the initial motion.

SECTION C | Q5 - OR: Centripetal force

OR - Define centripetal force, give its formula and explain its provision for the Moon. [1 + 1 + 3]

Centripetal force is the resultant force directed towards the centre of a circular path that keeps a body moving along that path.

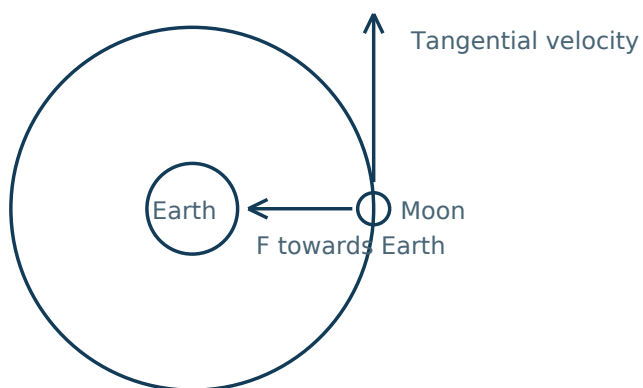
Formula: $F_c = mv^2/r$, where m is mass, v is speed and r is the radius of the circular path.

In the Moon's case: Earth's gravitational attraction supplies the centripetal force. It acts towards Earth's centre and continually changes the direction of the Moon's velocity, keeping it in orbit. In the circular approximation:

$$GM_E m_M / r^2 = m_M v^2 / r.$$

Gravity itself is the centripetal force; they are not two separate forces to be added.

Gravity supplies the centripetal force



SECTION C | Q6 - Thermometers [2.5 + 2.5]

Describe the construction and working of a liquid-in-glass thermometer and a thermocouple thermometer.

A. Liquid-in-glass thermometer

Construction: It consists of a glass bulb connected to a narrow, uniform capillary tube. The bulb and part of the tube contain a thermometric liquid, such as coloured alcohol or mercury. The tube is sealed and a calibrated temperature scale is marked beside it.

Working: When the bulb is placed in contact with a warmer body, the liquid gains heat and expands more than the glass. It rises in the narrow tube. When it cools, it contracts and the level falls. After the thermometer reaches thermal equilibrium with the object, temperature is read from the liquid level against the scale.

B. Thermocouple thermometer

Construction: It uses two wires of different metals, for example copper and constantan, joined to form a sensing junction. The other ends are connected to a sensitive voltmeter or measuring circuit. A reference junction is maintained at a known temperature, or its temperature is measured and electronically compensated.

Working: When the sensing junction and the reference junction are at different temperatures, a small thermoelectric voltage is produced by the **Seebeck effect**. This voltage depends on the temperature difference. The calibrated measuring instrument uses it, together with the known reference temperature, to indicate the sensing-junction temperature. Its small junction responds quickly and is useful over a wide temperature range.

Key distinction: The liquid thermometer uses thermal expansion; the thermocouple uses a thermoelectric voltage. A thermocouple measures temperature through its voltage, not through expansion of liquid.

SECTION C | Q6 - OR: Induced magnetism [1 + 2 + 2]

Define induced magnetism and briefly state any two ways to make an induced magnet.

Induced magnetism is the magnetism acquired by an initially unmagnetised magnetic material when it is placed in an external magnetic field. The field tends to align its magnetic domains. Soft iron is suitable for making a temporary induced magnet.

Method 1: Using a permanent magnet.

Bring a strong permanent magnet close to a soft-iron bar, or place the bar in contact with a pole. The magnet's field magnetises the bar by induction. The end nearer the magnet becomes the **opposite pole**, while the farther end becomes the same pole. For example, if the north pole is near the bar, the near end becomes south and the far end becomes north.



The end nearer the N pole becomes S; the farther end becomes N.

Method 2: Using a current-carrying coil.

Place a soft-iron bar inside a coil of insulated wire (a solenoid) and pass direct current through the coil. The coil produces a magnetic field that magnetises the iron, forming an electromagnet. When the current is switched off, soft iron loses most of its induced magnetism. Reversing the current reverses the poles.

Remember: The cause is an external magnetic field in both methods. Soft iron loses most of its magnetism when the inducing field is removed; steel can retain much more magnetism.

Final numerical answers for rapid revision

Question	Answer
Q4 - dropped block	$t = 8 \text{ s}$; $h = 320 \text{ m}$
Q4 OR - hydraulic jack	$F_1 \approx 1.67 \times 10^3 \text{ N}$
Q5 - car	Stopping-force magnitude = 5000 N
Q5 - van	Stopping-force magnitude = 4000 N
Q5 comparison	Car requires the greater stopping force

Exam presentation: Number each answer correctly, show formulas and units, box final results, and label diagrams.